Felt Shrinkage Rate Worksheet

Wool typically shrinks ~30% through the felting process. However, this can change depending on the variety of wool and the level of fulling.

Understanding shrinkage rate is extremely important when making objects of a specified size. For instance, for a slipper to fit your foot, your starting template should be much larger (see image). Shrinkage rate can vary between the length and width, so calculate separately for the most precise results.

Following are two key formulas. Fill in the tables to calculate the correct template size for your Shoebox Object #2: Flat Felt Carpet.

Formula #1

Shrinkage Rate = change in size of sample ÷ starting size of sample

Breaking it down…

SHRINKAGE RATE: LENGTH		
A	**Starting length of sample** *the length of the foam template used in class*	
B	**Finished length of sample** *the length of your finished felt sample*	
C	**Change in length of sample** *starting length (A) minus finished length (B)*	
D	**Shrinkage rate of length** *change in size of sample (C) ÷ starting size of sample (A)*	
SHRINKAGE RATE: WIDTH		
E	**Starting width of sample** *the length of the foam template used in class*	
F	**Finished width of sample** *the length of your finished felt sample*	
G	**Change in width of sample** *starting width (A) minus finished width (B)*	
H	**Shrinkage rate of width** *change in size of sample (C) ÷ starting size of sample (A)*	

The shrinkage rate will be a decimal number lower than 1 (e.g. 0.3). Use this number for formula #2 overleaf. To describe the shrinkage rate, you can read it as a percentage of 1. For example, 0.3 is 30%.

Now you have calculated the shrinkage rate, you can work out the template size you need to create a flat sheet of a specific size.

Formula #2

Template size = target size ÷ (1 – shrinkage rate)

Breaking it down…

TEMPLATE SIZE: LENGTH		
I	**Target length** *Length of shoebox*	
J	**Template length** *Target length (I) ÷ (1 - shrinkage rate of length (D))*	
TEMPLATE SIZE: WIDTH		
K	**Target width** *Width of shoebox*	
L	**Template width** *Target width (K) ÷ (1 - shrinkage rate of width (H))*	